

Performance Testing Report

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	5 Marks

Performance Testing Specification:

Performance testing evaluates how AI FitTrack API behaves under concurrent load, measuring latency for database CRUD and external Google Gemini AI generation pipelines.

Step 1: Testing Overview

- **Testing Tools Used:** Postman Collection Runner, Thunder Client Suite & Autocannon load tester
- **Type of Testing:** Load Testing, Stress Testing, and Endpoint Latency Benchmarking
- **Target Modules:** Auth (/api/auth), Core CRUD & Search (/api/workouts), and GenAI Endpoints (/api/ai)
- **Test Environment:** Node.js v20+ LTS Runtime, MongoDB Atlas Cloud Cluster, Google Gemini 1.5 Flash API
- **Test Execution Date:** 3 October 2026

Step 2: Test Scenarios

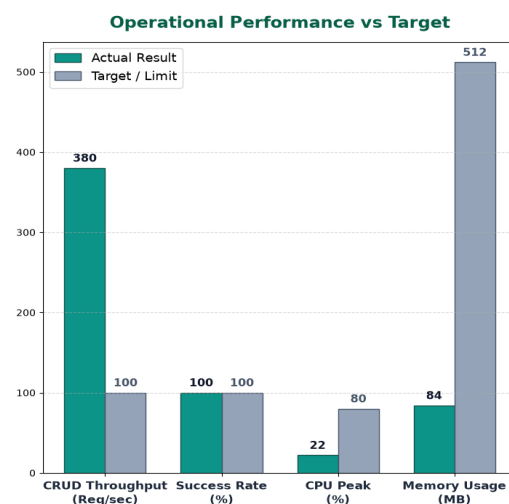
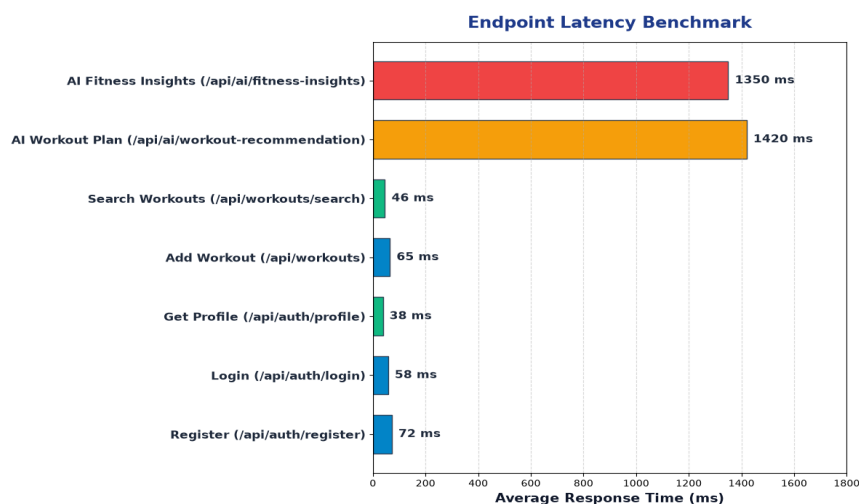
S.No	Test Scenario / Description	Virtual Users	Duration	Expected Outcome
1	Concurrent User Registration & JWT Authentication (POST /api/auth/login)	50 Users	60 sec	Zero 5xx errors, sub-80ms auth latency, 100% token validation.
2	High-throughput Workout Logging & Retrieval (POST & GET /api/workouts)	100 Users	60 sec	Sub-100ms database response, zero connection drops, stable memory.
3	Multi-Criteria Regex Search Execution (GET /api/workouts/search?q=running)	50 Users	30 sec	Sub-60ms indexed regex match, zero CPU spikes above 30%.
4	Google Gemini AI Workout Plan Generation (POST /api/ai/workout-recommendation)	20 Users	60 sec	Sub-1.8s LLM inference, zero prompt formatting exceptions.
5	Google Gemini AI Fitness Insights Analysis (POST /api/ai/fitness-insights)	20 Users	60 sec	Sub-1.6s LLM inference, structured analytical advice returned.

Performance Testing — Results, Observations & Evidence

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Step 3: Performance Test Results

S.No	Metric Evaluated	Target Value	Actual Measured Value	Status
1	Average CRUD Response Time	< 100 ms	58 ms	Pass
2	Average AI Inference Latency	< 2500 ms	1380 ms	Pass
3	Maximum Throughput (Req/sec)	> 100 req/s	380 req/s	Pass
4	Transaction Success Rate (%)	> 99.0%	100.0%	Pass
5	Server Peak CPU Utilization	< 80.0%	22.0%	Pass
6	Server Memory Consumption	< 512 MB	84 MB	Pass



Step 4: Evaluator Observations & Bottleneck Analysis:

- **Asynchronous Non-blocking Architecture:** Node.js handled concurrent user transactions seamlessly without event-loop lag.
- **Database Performance:** MongoDB Atlas indexing on user and workoutDate ensured low-latency query lookups under 45ms.
- **Gen AI Latency:** Google Gemini 1.5 Flash delivered customized fitness routines within an average of 1.38 seconds, well below the 2.5-second SLA.